

F78PB Probability and Statistics B

Moments and moment generating functions

Frequentist Set-up

We consider a repeatable experiment with probability model $(\Omega, \mathcal{F}, \mathbb{P})$ and outcome X.

Probability: Distribution known $\Rightarrow$ compute $\mathbb{P}, \mathbb{E}, \mathbb{V}\text{ar}, \ldots$

Statistics: Distribution unknown $\Rightarrow$ estimate location, spread, dependence.

Data: n i.i.d. copies $\underline{X} = (X_1, \ldots, X_n)$ observed as $\underline{x} = (x_1, \ldots, x_n)$.

Sample Moments

Random experiment X with n observations $\underline{x} = (x_1, \ldots, x_n)$.
General formula:

$$\mathbb{E}[(X - c)^k] \approx \frac{1}{n} \sum_{i=1}^{n} (x_i - c)^k.$$

Simple moments ($c = 0$):

$$\nu_k = \frac{1}{n} \sum_{i=1}^{n} x_i^k.$$

Central moments ($c = \bar{x}$):

$$m_k = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^k.$$

Special cases:

$$\bar{x} = \nu_1, \quad s^2 = \frac{n}{n-1} \mu_2.$$

Pearson's Symmetry Coefficient

If X symmetric $\Rightarrow$ odd central moments vanish.

$$\mathbb{E}[(X - \mathbb{E}[X])^k] = 0, \quad k = 1, 3, 5, \ldots$$

The first non-trivial choice is $k = 3$

$$\mathbb{E}[(X - \mathbb{E}[X])^3] = 0.$$

Scale with σ^3 to obtain a dimensionless quantity

$$\frac{\mathbb{E}[(X - \mathbb{E}[X])^3]}{\sigma^3}, \quad \sigma = \sqrt{\mathbb{V}\mathrm{ar}(X)}.$$

Estimator:

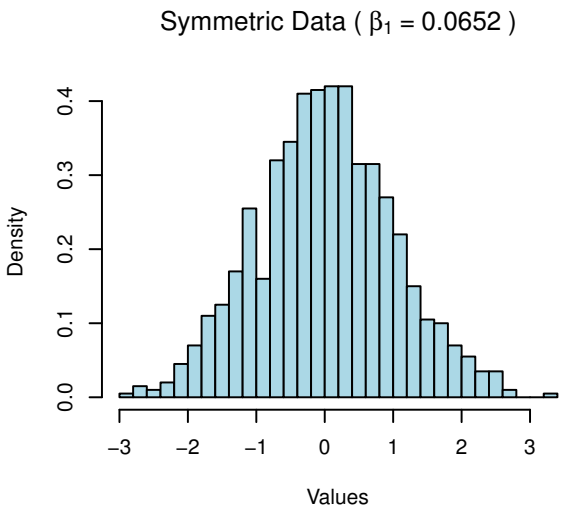
$$\beta_1 = \frac{m_3}{s^3}.$$

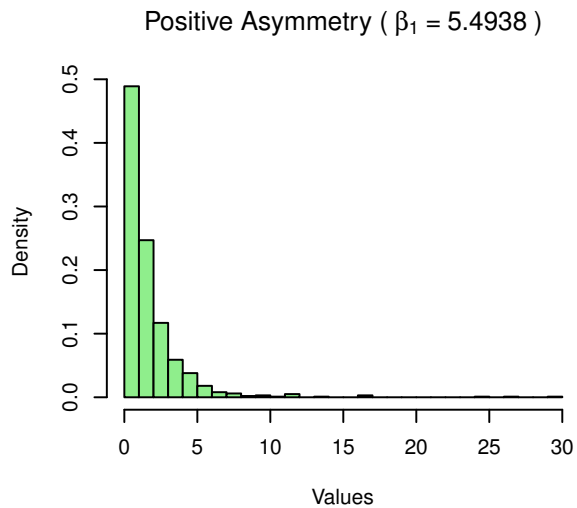
Interpretation:

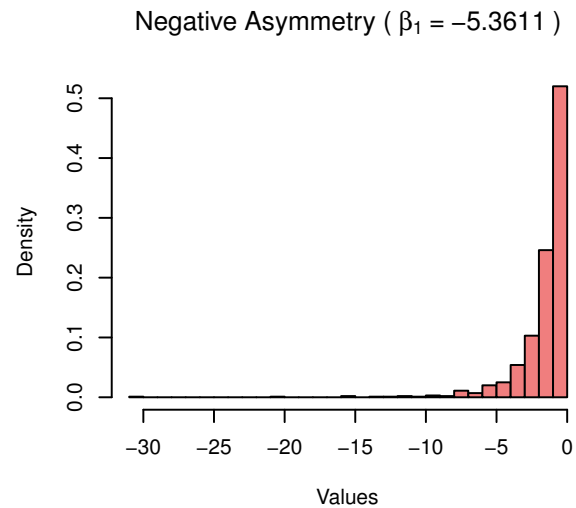
- $\beta_1 \approx 0$ symmetric
- $\beta_1 > 0$ right tail
- $\beta_1 < 0$ left tail

Warning: The converse does not necessarily hold!

Illustration



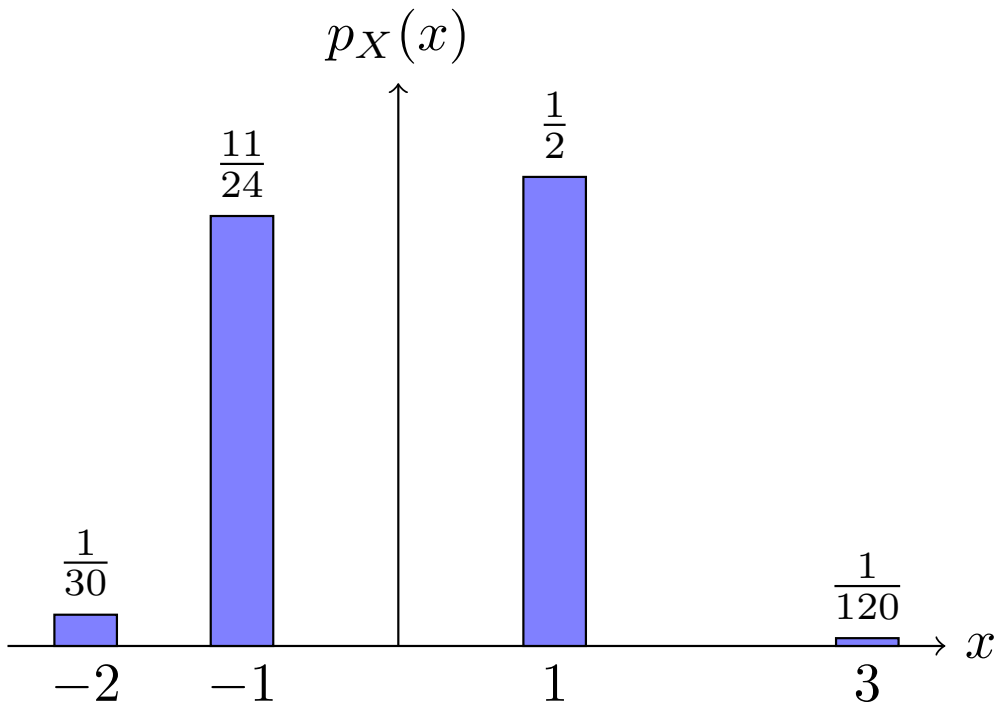




Example: $\beta_1 = 0$ does not imply symmetry

X with $\mathrm{Im}(X) = \{-2, -1, 1, 3\}$ and

$$p_X(-2) = \tfrac{1}{30},\ p_X(-1) = \tfrac{11}{24},\ p_X(1) = \tfrac{1}{2},\ p_X(3) = \tfrac{1}{120}.$$



Compute:

$$\mathbb{E}[X] = 0,\ \mathbb{E}[X^3] = 0,\ \mathbb{E}[X^5] = 1,\ \mathbb{V}\mathrm{ar}(X) = \tfrac{7}{6}.$$

So $\frac{\mathbb{E}[(X-\mathbb{E}X)^3]}{\sigma^3} = 0$ but distribution not symmetric.

Car Accidents

Example

The number of car accidents per day that occurred during November in a particular area are given in Table 1.

Accidents per day	0	1	2	3	4	5
Days (frequency)	2	4	5	9	7	3

Table: Car accidents per day

Calculate the coefficient of symmetry β_1.

Car Accidents Example

	0	1	2	3	4	5	Total	Notes
Accidents/day	0	1	2	3	4	5		y_i
Days (freq.)	2	4	5	9	7	3	30	f_i
Accid. $\times$ freq.	0	4	10	18	28	15	84	$y_i f_i$
Accid.$^2\times$ freq.	0	4	20	54	112	75	292	$y_i^2 f_i$
Accid.$^3\times$ freq.	0	4	40	162	448	375	1110	$y_i^3 f_i$

$$n = \sum f_i, \quad \bar{x} = \frac{\sum y_i f_i}{n},$$

$$\nu_2 = \frac{\sum y_i^2 f_i}{n}, \; s^2 = \frac{n}{n-1}(\nu_2 - \bar{x}^2),$$

$$\nu_3 = \frac{\sum y_i^3 f_i}{n}, \; m_3 = \nu_3 - 3\bar{x}\nu_2 + 3\bar{x}^2\nu_1 - \bar{x}^3.$$

$$n = 30, \; \bar{x} = 2.8,$$

$$\nu_2 = 292/30, \; s^2 \approx 1.959,$$

$$\nu_3 = 1110/30 = 37,$$

$$m_3 \approx -0.856.$$

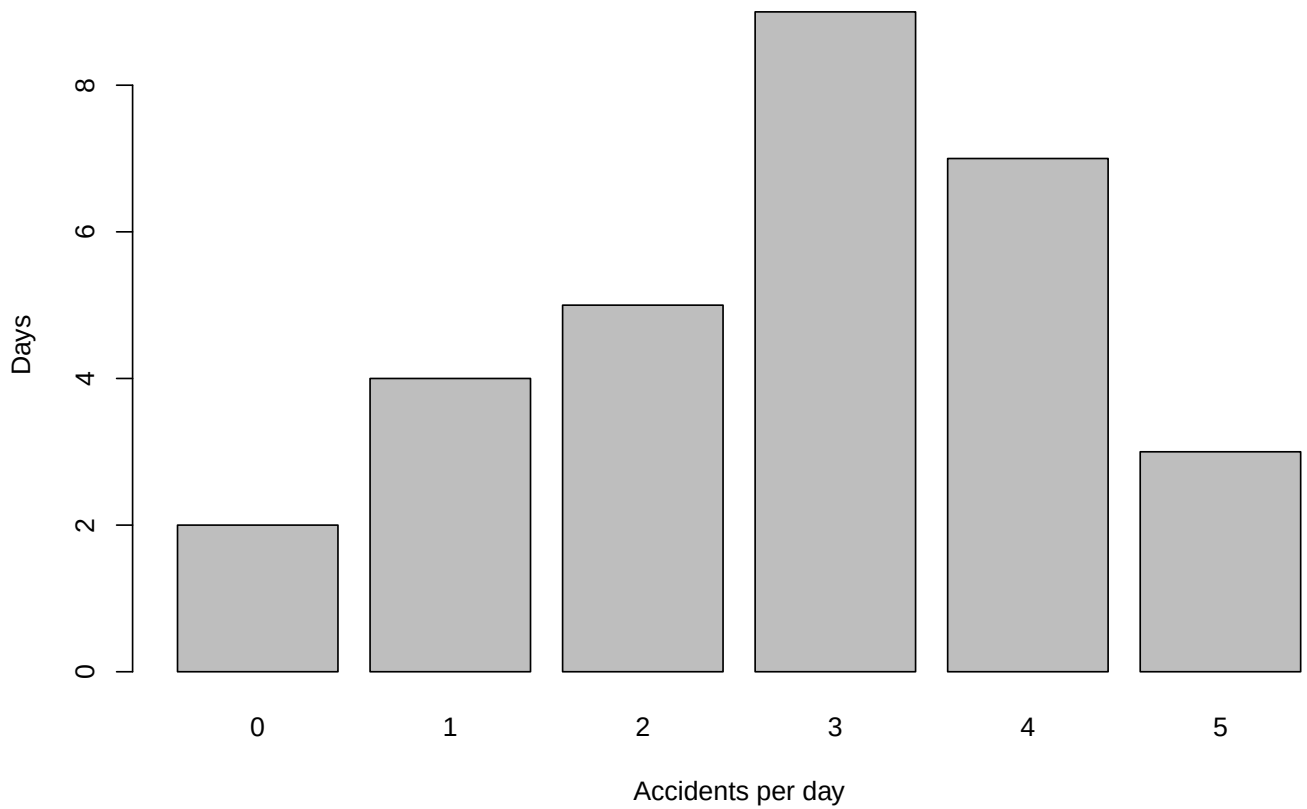
$$\beta_1 = \frac{m_3}{s^3} \approx \frac{-0.856}{(1.959)^{3/2}} \approx -0.312.$$

Data example: accidents per day

Example

Counts for $y = \{0, 1, 2, 3, 4, 5\}$ are $f = \{2, 4, 5, 9, 7, 3\}$.

$$\bar{x} = 2.8, \quad s^2 \approx 1.959, \quad \mu_3 \approx -0.856 \Rightarrow \beta_1 \approx -0.312.$$



Pearson's Kurtosis coefficient

$$\gamma_2 = \frac{\mu_4}{s^4} - 3 \;\approx\; \mathbb{E}\left[\left(\frac{X - \mathbb{E}X}{\sqrt{\mathbb{V}\mathrm{ar}(X)}}\right)^4\right] - 3.$$

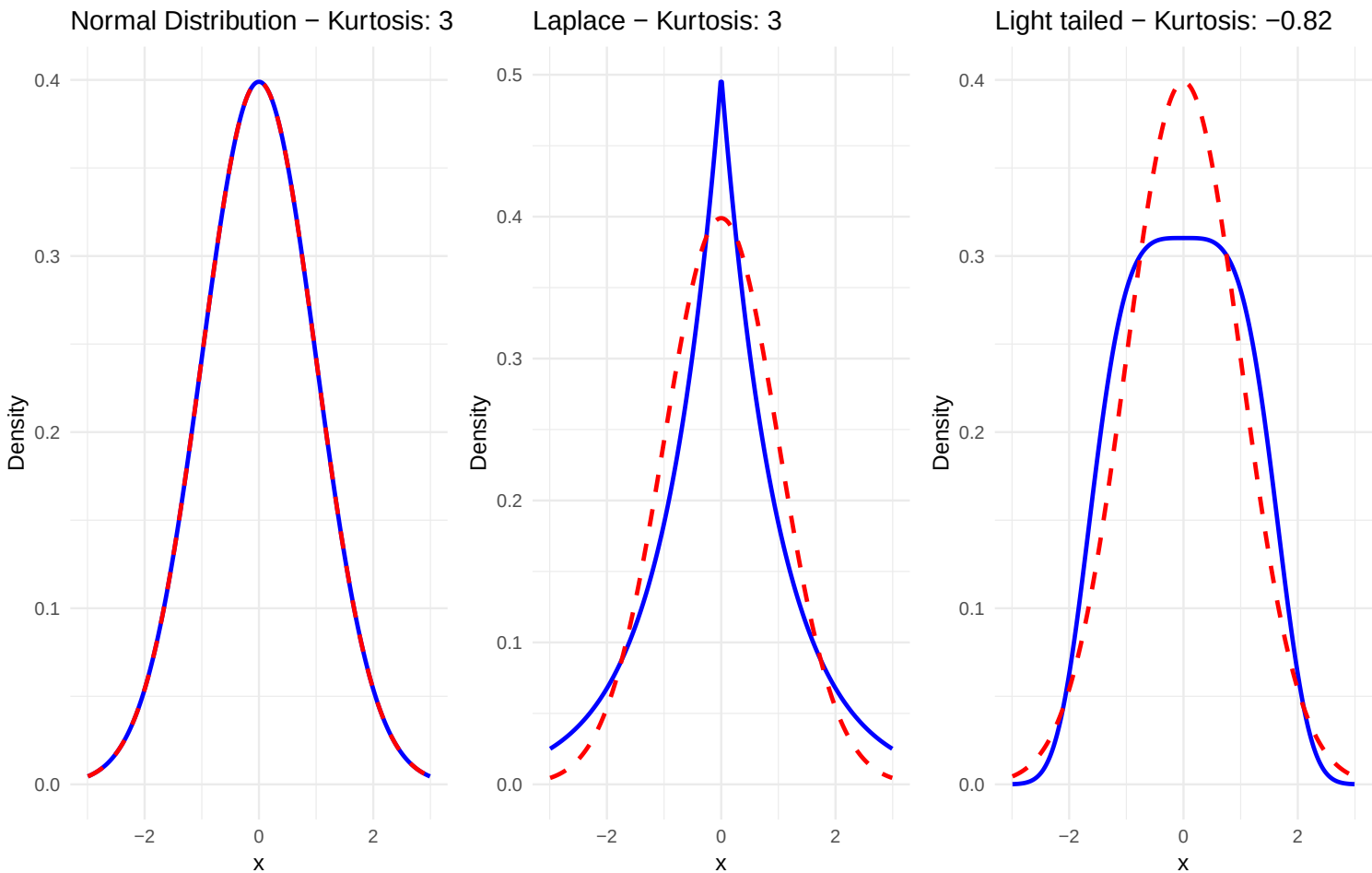
$\gamma_2 = 0$ (normal-like)

$\gamma_2 > 0$ (more peaked, heavier tails)

$\gamma_2 < 0$ (less peaked)

Warning: The converse does not necessarily hold!

Kurtosis


Normal Distribution − Kurtosis: 3
Laplace − Kurtosis: 3
Light tailed − Kurtosis: −0.82
Density
x
0.0
0.1
0.2
0.3
0.4
0.5
−2
0
2

Moment Generating Functions (mgf)

We define the moment generating function of a random variable X as

$$M_X(\theta) = \mathbb{E}(e^{\theta X}), \text{ for all } \theta \text{ where this expectation exists.}$$

For X a discrete random variable with probability mass function p_X:

$$M_X(\theta) = \sum_{x \in \mathrm{Im} X} e^{\theta x} p_X(x)$$

For X a continuous random variable with probability density function f_X:

$$M_X(\theta) = \int_{-\infty}^{\infty} e^{\theta x} f_X(x) \, \mathrm{d}x.$$

Example: $X \sim \text{Exp}(\lambda)$ ($f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$)

Definition:

$$M_X(\theta) = \int_0^\infty e^{\theta x} \lambda e^{-\lambda x} \, \mathrm{d}x.$$

Combine exponents:

$$M_X(\theta) = \int_0^\infty e^{\theta x} \lambda e^{-\lambda x} \, \mathrm{d}x = \lambda \int_0^\infty e^{(\theta-\lambda)x} \, \mathrm{d}x$$

The integral converges if and only if $\theta < \lambda$. Rewrite the integral as:

$$M_X(\theta) = \frac{\lambda}{\lambda - \theta} \int_0^\infty (\lambda - \theta) e^{(\theta-\lambda)x} \, \mathrm{d}x$$

$$M_X(\theta) = \frac{\lambda}{\lambda - \theta}, \qquad \text{for } \theta < \lambda$$

Result: $M_X(\theta) = \frac{\lambda}{\lambda-\theta}$ for $\theta < \lambda$.

Example: $Z \sim \mathrm{N}(0,1)$ ($f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$)

Definition:

$$M_Z(\theta) = \int_{-\infty}^{\infty} e^{\theta z} \frac{e^{-z^2/2}}{\sqrt{2\pi}} \, \mathrm{d}z.$$

Complete square:

$$\frac{z^2}{2} - \theta z = \frac{(z-\theta)^2}{2} - \frac{\theta^2}{2}.$$

Rearrange:

$$M_Z(\theta) = e^{\frac{\theta^2}{2}} \int_{-\infty}^{\infty} \underbrace{\frac{e^{-\frac{(z-\theta)^2}{2}}}{\sqrt{2\pi}}}_{\text{pdf of N}(\theta,1)} \, \mathrm{d}z.$$

Using that pdfs integrate to 1:

$$M_Z(\theta) = e^{\frac{\theta^2}{2}}.$$

Result: $M_X(\theta) = e^{\theta^2/2}$ for all $\theta \in \mathbb{R}$.

MGF formal calculations

Observe that

$$\frac{d}{d\theta}M_X(\theta) = \frac{d}{d\theta}\mathbb{E}[e^{\theta X}] = \mathbb{E}[\frac{d}{d\theta}e^{\theta X}] = \mathbb{E}[Xe^{\theta X}]$$

Now setting $\theta = 0$ we have

$$\frac{d}{d\theta}M_X(\theta)|_{\theta=0} = M'_X(0) = \mathbb{E}[X]$$

Repeating this with k derivatives

$$\frac{d^k}{d\theta^k}M_X(\theta)|_{\theta=0} = M_X^{(k)}(0) = \mathbb{E}[X^k]$$

Moments from Moment Generating Functions

$$\frac{d^k}{d\theta^k} M_X(\theta)|_{\theta=0} = M_X^{(k)}(0) = \mathbb{E}[X^k]$$

Using this calculation we can quickly calculate moments if we have the moment generating function.
An alternative approach is to use series expansions. Recall Taylor's series for a smooth function $g : \mathbb{R} \to \mathbb{R}$

$$g(\theta) = f(\theta_0) + f'(\theta_0)(\theta - \theta_0) + \frac{1}{2} f''(\theta_0)(\theta - \theta_0)^2 + \ldots = \sum_{k=0}^{\infty} \frac{f^{(k)}(\theta_0)(\theta - \theta_0)^k}{k!}.$$

The Taylor series for M_X is

$$M_X(\theta) = 1 + \mathbb{E}[X]\theta + \frac{1}{2}\theta^2 + \ldots = \sum_{k=0}^{\infty} \frac{M_X^{(k)}(0)\theta^k}{k!}.$$

Uniqueness

Consider two random variables X and Y. When do X and Y have the same distribution?

There exist random variables such that $\mathbb{E}[X^k] = \mathbb{E}[Y^k]$ for all $k = 0, 1, \ldots, n$ but X has a different distribution from Y.

If there exists $\delta > 0$ such that $M_X(\theta) = M_Y(\theta)$ for all $\theta \in (-\delta, \delta)$ then X and Y have the same distribution.

Linear Transformations of MGF

Let a, b be constants and X be a random variable then

$$M_{aX+b}(\theta) = \mathbb{E}\left[e^{(aX+b)\theta}\right] = e^b \mathbb{E}[e^{a\theta X}] = e^b M_X(a\theta).$$

If X and Y are independent then

$$M_{X+Y}(\theta) = \mathbb{E}[e^{\theta(X+Y)}] = \mathbb{E}[e^{\theta X} e^{\theta Y}] \overset{\text{indep.}}{=} \mathbb{E}[e^{\theta X}]\mathbb{E}[e^{\theta Y}] = M_X(\theta) M_Y(\theta).$$

It follows that for $X_1, \ldots, X_n$ independent and identically distributed copies of X

$$S = \sum_{i=1}^{n} X_i, \qquad M_S(\theta) = \prod_{i=1}^{n} M_{X_i}(\theta) = M_X(\theta)^n$$

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad M_{\bar{X}}(\theta) = \prod_{i=1}^{n} M_{X_i}\left(\frac{\theta}{n}\right) = M_X\left(\frac{\theta}{n}\right)^n.$$

Applications

Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim} \text{Exp}(\lambda)$ and let $S = \sum_{i=1}^{n} X_i$.
Recall

$$M_X(\theta) = \frac{\lambda}{\lambda - \theta} \quad \text{for } \theta < \lambda.$$

Using the property:

$$S = \sum_{i=1}^{n} X_i, \qquad M_S(\theta) = \prod_{i=1}^{n} M_{X_i}(\theta) = M_X(\theta)^n$$

gives

$$M_S(\theta) = M_X(\theta)^n = \left(\frac{\lambda}{\lambda - \theta}\right)^n.$$

Exercise: Show that the moment generating function of $Y \sim \text{Gamma}(\alpha, \lambda)$ is

$$M_Y(\theta) = \left(\frac{\lambda}{\lambda - \theta}\right)^\alpha.$$

Then by uniqueness of the moment generating function we have $S \sim \text{Gamma}(n, \lambda)$.

Probability Generating Function

If X is a random variable which takes non-negative integer values ($\mathrm{Im}(X) \subseteq \{0, 1, 2, \ldots\}$) then it is useful to work with the *probability generating function* G_X for X:

$$G_X(s) = \mathbb{E}(s^X) = \sum_{k=0}^{\infty} \mathbb{P}(X = k) s^k.$$

Binomial Distribution

Recall the Binomial Theorem:

$$(a+b)^n = \sum_{k=0}^{n} a^k b^{n-k}.$$

For $X \sim \text{Bin}(n,p)$ we have

$$\begin{aligned} G_X(s) &= \mathbb{E}\left[s^X\right] \\ &= \sum_{k=0}^{\infty} \mathbb{P}(X=k)s^k \\ &= \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} s^k \\ &= (sp + (1-p))^n. \end{aligned}$$

Geometric Distribution

Recall the geometric series:

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r} \quad \text{for } |r| < 1.$$

For $X \sim \text{Geo}(p)$ we have

$$\begin{aligned} G_X(s) &= \mathbb{E}[s^X] \\ &= \sum_{k=0}^{\infty} \mathbb{P}(X = k) s^k \\ &= \sum_{k=1}^{\infty} p(1-p)^{k-1} s^k \\ &= ps \sum_{k=1}^{\infty} (s(1-p))^{k-1} \\ &= \frac{ps}{1-s(1-p)}. \end{aligned}$$

PGF Properties

Recall the definition

$$G_X(s) = \mathbb{E}[s^X] = \sum_{k=0}^{\infty} \mathbb{P}(X = k) s^k.$$

we derive:

$$\begin{aligned} G_X(0) &= \mathbb{P}(X = 0) \\ G_X(1) &= \mathbb{E}[1^X] = 1 \\ G_X^{(m)}(0) &= m!\, \mathbb{P}(X = m) \\ G_X^{(m)}(1) &= \mathbb{E}[X(X-1)\dots(X-m+1)] \end{aligned}$$

PGF Properties

Recall the definition

$$G_X(s) = \mathbb{E}[s^X] = \sum_{k=0}^{\infty} \mathbb{P}(X = k)s^k.$$

If X and Y are discrete random variables with

$$G_X(s) = G_Y(s) \quad \text{for all } s$$

then X and Y have the same distribution.

PGF Properties

From the definition

$$G_X(s) = \mathbb{E}[s^X] = \sum_{k=0}^{\infty} \mathbb{P}(X = k)s^k$$

For $X_1, \ldots, X_n$ independent discrete random variables, let $Z = \sum_{i=1}^{n} X_i$

$$\begin{aligned} G_Z(s) &= \mathbb{E}[s^{X_1+X_2+\ldots+X_n}] \\ &= \mathbb{E}[s^{X_1} s^{X_2} \ldots s^{X_n}] \\ &\overset{\text{independence}}{=} \mathbb{E}[s^{X_1}]\mathbb{E}[s^{X_2}] \ldots \mathbb{E}[s^{X_n}] \\ &= G_{X_1}(s)G_{X_2}(s) \ldots G_{X_n}(s) \\ &= \prod_{i=1}^{n} G_{X_i}(s). \end{aligned}$$

Compound Random Variables

Let N be a non-negative integer valued random variable.
Let $X_1, X_2, \ldots$ be a sequence of independent, identically distributed random variables with moment generating function M_X. Assume N is independent of $X_1, X_2 \ldots$.
If $S_N = X_1 + \cdots + X_N$:

$$\begin{aligned} M_{S_N}(\theta) &= \mathbb{E}[e^{\theta S_N}] \\ &= \sum_{n=0}^{\infty} \mathbb{E}[e^{\theta S_N} | N = n] \mathbb{P}(N = n) \\ &= \sum_{n=0}^{\infty} \mathbb{E}[e^{\theta S_n}] \mathbb{P}(N = n) \\ &= \sum_{n=0}^{\infty} M_X(\theta)^n \mathbb{P}(N = n) \\ &= G_N(M_X(\theta)) \end{aligned}$$

Cumulant Generating Function

The cumulant generating function K_X for X is defined by

$$K_X(\theta) = \log M_X(\theta) \quad \text{for all } \theta \text{ for which this expectation exists}$$

Define the cumulant κ_n as

$$\kappa_n = K_X^{(n)}(0).$$

Observe

$$\kappa_1 = K_X'(0) = \frac{M_X'(0)}{M_X(0)} = \mathbb{E}[X].$$

$$\kappa_2 = K_X''(0) = \frac{M_X(0)M_X''(0) - M_X'(0)^2}{M_X(0)^2} = \mathbb{V}\text{ar}(X).$$

Normal: for $X \sim \mathrm{N}(0,1)$ $K_X(\theta) = \mu\theta + \frac{\sigma^2\theta^2}{2} \Rightarrow \kappa_n = 0$ for $n \geq 3$.

Independent sums

For $X_1, \ldots, X_n$ independent we have seen

$$M_{X_1+\ldots+X_n}(\theta) = \prod_{i=1}^{n} M_{X_i}(\theta)$$

$$G_{X_1+\ldots+X_n}(s) = \prod_{i=1}^{n} G_{X_i}(s)$$

For the cumulant generating function we have

$$\begin{aligned} K_{X_1+\ldots+X_n}(\theta) &= \log\left(M_{X_1+\ldots+X_n}(\theta)\right) \\ &= \log\left(\prod_{i=1}^{n} M_{X_i}(\theta)\right) \\ &= \sum_{i=1}^{n} \log(M_{X_i}(\theta)) \\ &= \sum_{i=1}^{n} K_{X_i}(\theta). \end{aligned}$$